

C683

Natural Science Lab

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A student who keeps a garden journal and grows tomatoes on a south-facing balcony wants to test whether coffee grounds help plants grow. Her lab partner, who took chemistry last year, suggests several wordings. Which is a testable hypothesis?

- A. Tomatoes on a south-facing balcony grow better than tomatoes kept indoors.
- B. Coffee grounds are good for plants.
- C. Bean plants given 10 g of coffee grounds per week will grow taller over 4 weeks than bean plants given none.
- D. Coffee grounds will always make every kind of plant grow taller.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A lab group tests how water temperature affects how fast a sugar cube dissolves. They use water at 10°C, 30°C, and 50°C, time each cube with a phone stopwatch, and use the same brand of sugar cubes bought at a local grocery. What is the dependent variable?

- A. The water temperature.
- B. The time it takes each cube to dissolve.
- C. The brand of sugar cube.
- D. The phone stopwatch.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

A study's dependent variable is the number of seeds that sprout in each tray. Which of these could be the independent variable?

- A. The number of seeds that sprout.
- B. Using the same seed variety in every tray.
- C. The researcher's prediction about how many seeds will sprout.
- D. Soil temperature, set at 15°C, 20°C, and 25°C in different trays.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A class tests whether a new fertilizer increases radish mass. They grow 40 radishes in a greenhouse donated by a parent: 20 get the fertilizer and 20 get none, with the same soil, water, and light. The school's science fair is in May. What is the role of the 20 radishes that get no fertilizer?

- A. They are unnecessary, because the fertilized radishes alone show whether growth happened.
- B. They are the independent variable.
- C. They are the control group, giving a baseline to compare against.
- D. They are a controlled variable.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A student wants to know whether Brand A paper towels absorb more water than Brand B. She tests one sheet of each brand one time, and Brand A absorbs 2 mL more. The towels were on sale at the same store. What is the biggest weakness in concluding that Brand A absorbs more?

- A. The towels were bought on sale, so their quality may differ.
- B. There was only one trial per brand, so the 2 mL difference could be chance.
- C. There is no weakness; any measured difference proves Brand A always absorbs more.
- D. The dependent variable should have been price per sheet.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A team records reaction times in milliseconds for five trials using a free app on a school tablet: 250, 260, 255, 265, and 520. On the fifth trial the participant sneezed. Which measure of center best describes a typical trial, and what is its value?

- A. The median, 260 ms.
- B. 520 ms.
- C. The range, 270 ms.
- D. The mean, 310 ms.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Four trials have a mean of 20.0 seconds. Three of the times are 18, 21, and 19 seconds. What was the fourth time?

- A.** 22 seconds
- B.** 14.5 seconds
- C.** 19.3 seconds
- D.** 20 seconds

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A group measures the boiling point of pure water at sea level five times with a new digital thermometer: 96.1, 96.0, 96.2, 96.1, and 96.1°C. The accepted value is 100.0°C. How should the measurements be described?

- A.** Accurate, because the thermometer is new and digital.
- B.** Both accurate and precise, because the readings agree so closely.
- C.** Accurate but not precise.
- D.** Precise but not accurate.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

A student's measured density of aluminum has a percent error of 5%, and the measurement was below the accepted value of 2.70 g/cm³. What value did the student measure?

- A.** 2.65 g/cm³
- B.** 2.70 g/cm³
- C.** 2.57 g/cm³
- D.** 2.84 g/cm³

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

Researchers in a coastal town tracked ten summers and found that months with higher ice-cream sales also had more swimming accidents. The town's lifeguard program won a state award. What is the best conclusion?

- A. The two are correlated, likely through a third factor such as hot weather; the data do not show that ice cream causes accidents.
- B. Eating ice cream causes swimming accidents.
- C. The lifeguard program must be failing.
- D. Swimming accidents cause people to buy more ice cream.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

Which study design best supports a claim that a vitamin supplement causes faster wound healing?

- A. Survey people who already take the supplement and compare their healing to non-users.
- B. Give the supplement to one volunteer and watch the wound heal.
- C. Find a strong correlation ($r = 0.9$) in a large observational data set.
- D. Randomly assign patients to the supplement or a placebo and compare healing times.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

For the research part of the lab, a student needs evidence about ocean acidification. She has a 2-hour block in the library on Thursday and prefers sources with lots of charts. Which source is most appropriate as primary academic evidence?

- A. A personal blog post with many colorful charts.
- B. A peer-reviewed journal article reporting original measurements of seawater pH.
- C. Any encyclopedia entry, since everything in print is reliable.
- D. A popular science magazine's summary of that study.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A student ~~who keeps a garden journal and grows tomatoes on a south facing balcony~~ wants to test whether coffee grounds help plants grow. ~~Her lab partner, who took chemistry last year, suggests several wordings.~~ Which is a testable hypothesis?

Target word: which wording is a testable hypothesis **Decides:** Only one option names a measurable treatment, a measurable outcome, and a comparison.

A	Camouflage	Answers the balcony detail and drops the coffee-grounds question entirely.
B	Dropped element	No measurable outcome or comparison; “good” cannot be tested as stated.
C	Correct	Specific treatment, measurable outcome (height), a comparison group, and it can be proven wrong.
D	Absolute language	A universal claim about every plant cannot be supported by one bounded experiment.

2. Answer B Concept 2

A lab group tests how water temperature affects how fast a sugar cube dissolves. They use water at 10°C, 30°C, and 50°C, time each cube ~~with a phone stopwatch~~, and use the same brand of sugar cubes ~~bought at a local grocery~~. What is the dependent variable?

Target word: the dependent variable **Decides:** Dissolving time is what is measured in response to the temperature change.

A	Inverted operation	Temperature is changed on purpose, so it is the independent variable.
B	Correct	It is the outcome measured in response to the change the group makes.
C	Adjacent category	The brand is held the same for all trials, so it is a controlled variable.
D	Camouflage	The stopwatch is the measuring tool, not a variable.

3. Answer D Concept 2 · transfer

A study's dependent variable is the number of seeds that sprout in each tray. Which of these could be the independent variable?

Target word: work backward from the dependent variable to the independent variable **Decides:** The independent variable is the condition deliberately set differently across trays.

A	Inverted operation	That is the dependent variable given in the stem.
B	Adjacent category	Something kept the same in every tray is a controlled variable.
C	Confusion set	That is the hypothesis, not a variable.
D	Correct	It is changed on purpose across trays and could affect sprouting.

4. Answer C Concept 3

A class tests whether a new fertilizer increases radish mass. They grow 40 radishes in a greenhouse ~~donated by a parent~~: 20 get the fertilizer and 20 get none, with the same soil, water, and light. ~~The school's science fair is in May.~~ What is the role of the 20 radishes that get no fertilizer?

Target word: role of the untreated group **Decides:** They receive none of the treatment and give a baseline for comparison.

A	Dropped element	Without a baseline you cannot tell whether growth came from the fertilizer.
B	Adjacent category	The independent variable is fertilizer (yes or no), not the group itself.
C	Correct	A group that gets no treatment shows what happens without it.
D	Confusion set	A controlled variable is a condition held constant (like soil or water), not a group of subjects.

5. Answer B Concept 4

A student wants to know whether Brand A paper towels absorb more water than Brand B. She tests one sheet of each brand one time, and Brand A absorbs 2 mL more. ~~The towels were on sale at the same store.~~ What is the biggest weakness in concluding that Brand A absorbs more?

Target word: main weakness of the conclusion **Decides:** Only one trial per brand.

A	Camouflage	Where or how cheaply they were bought does not decide absorbency results.
B	Correct	Repeated trials are needed to separate a real difference from normal variation.
C	Absolute language	One measurement cannot show that a difference always holds.
D	Adjacent category	Price is a different question; absorbency is the right measure for this claim.

6. Answer A Concept 5

A team records reaction times in milliseconds for five trials ~~using a free app on a school tablet~~: 250, 260, 255, 265, and 520. On the fifth trial the participant sneezed. Which measure of center best describes a typical trial, and what is its value?

Target word: best measure of center and its value **Decides:** The 520 ms outlier pulls the mean; the median resists it.

A	Correct	Sorted: 250, 255, 260, 265, 520; the middle value is 260 and is not pulled by the outlier.
B	Wrong stem number	That is the outlier itself, not a typical value.
C	Confusion set	Range (520 – 250) measures spread, not center.
D	Adjacent category	$1,550 / 5 = 310$, but the single outlier pulls it above every normal trial.

7. Answer A Concept 5 • transfer

Four trials have a mean of 20.0 seconds. Three of the times are 18, 21, and 19 seconds. What was the fourth time?

Target word: solve for a missing data value **Decides:** Total must be $4 \times 20.0 = 80$ seconds.

A	Correct	$80 - (18 + 21 + 19) = 80 - 58 = 22$.
B	Inverted operation	Divides the known sum (58) by 4 instead of subtracting it from 80.
C	Dropped element	Averages the three known times and ignores the required total.
D	Wrong stem number	That is the mean given in the stem, not the missing value.

8. Answer D Concept 6

A group measures the boiling point of pure water at sea level five times ~~with a new digital thermometer~~: 96.1, 96.0, 96.2, 96.1, and 96.1°C. The accepted value is 100.0°C. How should the measurements be described?

Target word: accuracy vs. precision **Decides:** Readings cluster tightly but sit about 4°C below the accepted value.

A	Camouflage	A new instrument does not make readings accurate; the data show a consistent error.
B	Confusion set	Agreement between readings is precision; accuracy needs closeness to the accepted value.
C	Inverted operation	Reverses the two terms; the readings are tightly grouped and off target.
D	Correct	They agree closely with each other (precise) but are far from 100.0°C (not accurate).

9. Answer C Concept 6 • transfer

A student's measured density of aluminum has a percent error of 5%, and the measurement was below the accepted value of 2.70 g/cm³. What value did the student measure?

Target word: solve for the measured value from percent error **Decides:** Measured = $2.70 \times (1 - 0.05)$.

A	Confusion set	Treats 5% as 0.05 g/cm ³ instead of 5% of the accepted value.
B	Wrong stem number	That is the accepted value, not the measurement.
C	Correct	$2.70 \times 0.95 = 2.565$, about 2.57 g/cm ³ .
D	Inverted operation	2.70×1.05 ; the stem says the measurement was low.

10. Answer A Concept 7

Researchers in a coastal town tracked ten summers and found that months with higher ice-cream sales also had more swimming accidents. ~~The town's lifeguard program won a state award.~~ What is the best conclusion?

Target word: what the correlation supports **Decides:** Observational data with a likely third variable (hot weather).

A	Correct	Both rise in hot months; observational data cannot establish cause.
B	Confusion set	Reads correlation as causation.
C	Camouflage	The award detail does not bear on what the correlation shows.
D	Inverted operation	Reverses the direction but still assumes causation.

11. Answer D Concept 7 • transfer

Which study design best supports a claim that a vitamin supplement causes faster wound healing?

Target word: design that supports causation **Decides:** Random assignment plus a control group.

A	Adjacent category	Observational; users may differ in diet, age, or health.
B	Dropped element	No control group and no replication.
C	Confusion set	Even a strong correlation does not establish cause.
D	Correct	Randomization and a control group rule out other explanations.

12. Answer B Concept 8

For the research part of the lab, a student needs evidence about ocean acidification. ~~She has a 2-hour block in the library on Thursday and prefers sources with lots of charts.~~ Which source is most appropriate as primary academic evidence?

Target word: strongest academic source **Decides:** Peer-reviewed article reporting original data.

A	Camouflage	Charts are a preference, not a sign of credibility; blogs are unreviewed.
B	Correct	Primary, peer-reviewed research is the strongest academic evidence.
C	Absolute language	Being in print does not make a source reliable or primary.
D	Adjacent category	Secondary and not peer reviewed.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Testable hypothesis	A hypothesis is a specific, measurable, falsifiable prediction about how one variable affects another.	Accepting a vague opinion or a universal claim as a hypothesis.	Answer choices that differ in how measurable the wording is.
2	Independent vs. dependent variable	The independent variable is what the researcher changes; the dependent variable is what is measured in response.	Swapping the two, or calling a held-constant condition a variable of interest.	“affects,” “measured,” or “changed across groups.”
3	Control group and controlled variables	A control group receives no treatment and gives a baseline; controlled variables are conditions held the same for every group.	Treating “control group” and “controlled variable” as the same thing.	A group that gets none of the treatment.
4	Replication and sample size	Conclusions need repeated trials and adequate samples so chance variation can be separated from a real effect.	Trusting a single trial because it showed a difference.	“One sheet,” “once,” or a single measurement per group.
5	Mean vs. median with outliers	The mean is pulled by outliers; the median resists them and better shows a typical value in skewed data.	Always reporting the mean.	One value far from the others in a data list.
6	Accuracy vs. precision and percent error	Accuracy is closeness to the accepted value; precision is closeness of repeated measurements to each other. Percent error = $ \text{measured} - \text{accepted} / \text{accepted} \times 100$.	Calling tightly clustered readings accurate.	Repeated readings plus an accepted value.
7	Correlation vs. causation	A correlation shows variables move together; only a controlled experiment supports cause and effect, and a third variable may drive both.	Reading a strong correlation as proof of cause.	Observational data where two things rise together.
8	Credible academic sources	Peer-reviewed primary research is the strongest academic evidence; popular summaries and blogs are secondary or unreviewed.	Judging a source by its visuals or by being in print.	“Most appropriate academic source.”

Course scope (public description): This course provides students an introduction to using the scientific method and engaging in scientific research to reach conclusions about the natural world. Students will design and carry out an experiment in the natural sciences to investigate a hypothesis by gathering quantitative data. They will also research a specific topic in the natural sciences using academic sources and draw conclusions from their findings.